

[Aim: 100/100 in Maths]

आरंभ

POLYNOMIALS

Lecture #5

ATK²:

[variable]^{whole no}

poly.

$$ax^2 + bx + c$$

α

β

$$\checkmark \alpha + \beta = -\frac{b}{a}$$

$$\checkmark \alpha \cdot \beta = \frac{c}{a}$$

V.IMP

#LP: If ' α ' and ' β ' are the zeroes of the polynomial $p(y) = y^2 - 5y + 3$, then find the value of $\alpha^4\beta^3 + \alpha^3\beta^4$. (2025)

$$y^2 - 5y + 3$$

α
 β

$$\alpha + \beta = \frac{-(-5)}{1} = 5$$

$$\alpha \cdot \beta = \frac{3}{1} = 3$$

$$\Rightarrow \alpha^4\beta^3 + \alpha^3\beta^4$$
$$\Rightarrow \alpha^3\beta^3 [\alpha + \beta]$$

$$\Rightarrow (\alpha\beta)^3 [\alpha + \beta]$$

$$\Rightarrow (3)^3 [5]$$

$$\Rightarrow 27 \times 5$$

$$\Rightarrow \underline{135}$$

#LP: For what value of k , the product of zeroes of the polynomial $kx^2 - 4x - 7$ is 2 (2024)

(a) $-\frac{1}{14}$

✓ (b) $-\frac{7}{2}$

(c) $\frac{7}{2}$

(d) $-\frac{2}{7}$

given, $\alpha \cdot \beta = 2$

$$\Rightarrow \frac{c}{a} = 2$$

$$\Rightarrow \frac{-7}{k} = 2$$

$$\Rightarrow \boxed{\frac{-7}{2} = k}$$

#LP: If the sum of zeroes of the polynomial $p(x) = 2x^2 - k\sqrt{2}x + 1$ is $\sqrt{2}$, then value of k is: [CBSE 2024]

(A) $\sqrt{2}$

(B) 2 ✓

(C) $2\sqrt{2}$

(D) $\frac{1}{2}$

$$2x^2 - k\sqrt{2}x + 1$$

← α
← β

given, $\alpha + \beta = \sqrt{2}$

$$\frac{-b}{a} = \sqrt{2}$$

$$\frac{-(-k\sqrt{2})}{2} = \sqrt{2}$$

$$k = 2$$

K³B :

$$(\alpha + \beta)^2 = \alpha^2 + \beta^2 + 2\alpha\beta$$

$$\Rightarrow \boxed{\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta}$$

$$(\alpha + \beta)^3 = \alpha^3 + \beta^3 + 3\alpha\beta(\alpha + \beta)$$

$$\Rightarrow \boxed{\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)}$$

Q: If α & β are zeroes of ax^2+bx+c find the value of ~~(i) $\alpha-\beta$~~ ~~(ii) $\alpha^4+\beta^4$~~ ~~(iii) $\alpha^6+\beta^6$~~

Sol

$$\begin{aligned} & \cancel{ax^2+bx+c} \quad \swarrow \begin{matrix} \alpha \\ \beta \end{matrix} \\ & \cancel{\alpha+\beta = -\frac{b}{a}} \\ & \cancel{\alpha \cdot \beta = \frac{c}{a}} \end{aligned}$$

$$\begin{aligned} \text{(i)} \quad & \cancel{(\alpha-\beta)^2 = \alpha^2 + \beta^2 - 2\alpha\beta} \\ & \cancel{(\alpha+\beta)^2 = \alpha^2 + \beta^2 + 2\alpha\beta} \end{aligned}$$

$$(\alpha-\beta)^2 - (\alpha+\beta)^2 = -4\alpha\beta$$

$$\Rightarrow (\alpha-\beta)^2 = (\alpha+\beta)^2 - 4\alpha\beta$$

$$\Rightarrow \alpha-\beta = \sqrt{(\alpha+\beta)^2 - 4\alpha\beta}$$

$$\Rightarrow \alpha-\beta = \sqrt{\left(-\frac{b}{a}\right)^2 - 4\frac{c}{a}}$$

$$(ii) \alpha^4 + \beta^4$$

$$\Rightarrow (\alpha^2)^2 + (\beta^2)^2$$

$$\Rightarrow (\alpha^2 + \beta^2)^2 - 2\alpha^2\beta^2$$

$$\Rightarrow \left[(\alpha + \beta)^2 - 2\alpha\beta \right]^2 - 2(\alpha\beta)^2$$

$$\Rightarrow \left[\left(\frac{-b}{a}\right)^2 - 2\left(\frac{c}{a}\right) \right]^2 - 2\left(\frac{c}{a}\right)^2$$

$$\boxed{a^2 + b^2 = (a+b)^2 - 2ab}$$

$$(iii) \alpha^6 + \beta^6$$

$$(\alpha^3)^2 + (\beta^3)^2$$

$$\Rightarrow (a+b)^3 - 3ab(a+b)$$

$$\Rightarrow (\alpha^2 + \beta^2)^3 - 3\alpha^2\beta^2(\alpha^2 + \beta^2)$$

$$\Rightarrow \left[(\alpha + \beta)^2 - 2\alpha\beta \right]^3 - 3(\alpha\beta)^2 \left[(\alpha + \beta)^2 - 2\alpha\beta \right]$$

#LP: Find the value of k such that the polynomial $x^2 - (k + 6)x + 2(2k - 1)$ has sum of its zeroes equal to half of their product.

A7Q

$$SOR = \frac{1}{2} POR$$

$$\frac{-b}{a} = \frac{1}{2} \frac{c}{a}$$

$$-b = \frac{1}{2} \times c$$

$$-(-(k+6)) = \frac{1}{2} \times 2(2k-1)$$

$$k+6 = 2k-1$$

$$6+1 = 2k-k$$

$$\boxed{7 = k}$$

#LP: Find the value of p for which one root of the quadratic equation $px^2 - 14x + 8 = 0$ is 6 times the other.

$$px^2 - 14x + 8$$

α
 6α

$$\text{SOR} = -\frac{b}{a}$$

$$\Rightarrow \alpha + 6\alpha = -(-14)$$

$$\Rightarrow 7\alpha = \frac{14}{p}$$

$$\alpha = \frac{2}{p}$$

$$\text{POR} = \frac{c}{a}$$

$$\Rightarrow \alpha \cdot 6\alpha = \frac{8}{p}$$

$$\Rightarrow 6\alpha^2 = \frac{8}{p}$$

$$\Rightarrow 6\left(\frac{2}{p}\right)^2 = \frac{8}{p}$$

$$6 \times \frac{4}{p^2} = \frac{8}{p}$$

$$3 = p$$

#LP: If one zero of the quadratic polynomial $f(x) = 4x^2 - 8kx - 9$ is negative of the other, find the value of k .

$$4x^2 - 8kx - 9 \quad \begin{matrix} \alpha \\ -\alpha \end{matrix}$$

$$\text{SOR} = -\frac{b}{a}$$

$$(\alpha) + (-\alpha) = -\frac{(-8k)}{4}$$

$$0 = \frac{8k}{4}$$

$$\boxed{k=0}$$

$$\text{POR} = \frac{c}{a}$$

#LP: The zeroes of a polynomial $x^2 + px + q$ are twice the zeroes of the polynomial $4x^2 - 5x - 6$. The value of p is: [CBSE 2024]

(A) $-\frac{5}{2}$

(B) $\frac{5}{2}$

(C) -5

(D) 10

$$2\alpha + 2\beta = -\frac{p}{1}$$

$$2 \times \left(\frac{+5}{4} \right) = -\frac{p}{1}$$

$$\alpha = \frac{5}{2}$$

$$x^2 + px + q$$

$$2\alpha$$

$$2\beta$$

$$4x^2 - 5x - 6$$

$$\alpha + \beta = -\frac{(-5)}{4} \Rightarrow$$

$$\alpha + \beta = \frac{5}{4}$$

$$\alpha \cdot \beta = -\frac{6}{4} \Rightarrow$$

$$\alpha \cdot \beta = -\frac{3}{2}$$

#LP: If the zeroes of the polynomial $x^2 + ax + b$ are in the ratio 3:4, then prove that $12a^2 = 49b$.

$$\boxed{x^2 + ax + b} \begin{cases} 3k \\ 4k \end{cases}$$

Let the roots are $3k$ & $4k$

$$\text{SOR} = -\frac{b}{a}$$

$$3k + 4k = -\frac{a}{1}$$

$$7k = -a$$

$$\boxed{a = -7k}$$

$$\text{POR} = \frac{c}{a}$$

$$3k \cdot 4k = \frac{b}{1}$$

$$\boxed{12k^2 = b}$$

To prove:- $12a^2 = 49b$

LHS

$$12 \times a^2$$

$$\rightarrow 12 \times (-7k)^2$$

$$\text{LHS} \rightarrow 12 \times 49k^2$$

$$\text{LHS} = \text{RHS}$$

Hence proved

RHS

$$\rightarrow 49 \times b$$

$$\text{RHS} \rightarrow 49 \times 12k^2$$

Making of a Quadratic Polynomial:

To make a quad. poly, we only need 2 things.

SOR(s) & POR(P)

✓ ✓ Quad Poly $\Rightarrow K [x^2 - \underline{s}x + \underline{p}]$

my real no.
generally $K=1$

#LP: Find the quadratic polynomial whose zeros are ²-3 and ³-5.

$$S = \alpha + \beta$$
$$= -3 + (-5)$$

$$S = -8$$

$$P = \alpha \cdot \beta$$
$$= (-3)(-5)$$

$$P = 15$$

$$\text{Quad} \Rightarrow k(x^2 - Sx + P)$$

$$\Rightarrow k(x^2 - (-8)x + 15)$$

$$\text{Quad.} \Rightarrow \underline{k}(x^2 + 8x + 15)$$

$$k=1$$
$$\text{Quad} \Rightarrow [x^2 + 8x + 15]$$

$$k=2$$
$$\text{Quad} \Rightarrow 2[x^2 + 8x + 15]$$
$$\Rightarrow [2x^2 + 16x + 30]$$

$$k=3$$
$$k=4$$
$$k=5$$

K^3B

There are ∞ no. of quadratic polynomials possible with 2 roots.

#LP: Form a quadratic polynomial whose zeroes are $3 + \sqrt{2}$ and $3 - \sqrt{2}$. (2012)

$\swarrow$
 S
 $\searrow$
 P

$$S = (3 + \sqrt{2}) + (3 - \sqrt{2})$$

$$\boxed{S = 6} \checkmark$$

$$P = (3 + \sqrt{2})(3 - \sqrt{2})$$

$$P = (3)^2 - (\sqrt{2})^2$$

$$P = 9 - 2$$

$$\boxed{P = 7} \checkmark$$

α β

$$\underline{\underline{Q_{\text{quad}}}} \Rightarrow k [x^2 - Sx + P]$$

$$k [x^2 - 6x + 7]$$

$k=1 \rightarrow$

$$\boxed{x^2 - 6x + 7} \checkmark$$

#LP: Construct a quadratic polynomial, the sum and product of whose zeroes are

$$\underbrace{-\frac{1}{4}}_S \text{ and } \underbrace{\frac{1}{4}}_P$$

respectively. (CBSE 2017)



Full Complete Module

except

$\frac{1^2(1)}{2^2(1)} \frac{5, 16, 7}{8}$

THANK YOU COODIESS !!

